

D780

Software Architecture and Design

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An order module queries the billing module's internal database tables directly and reads 14 of its private fields. The code is eight years old and well commented. What is the main design problem?

- A. No problem, because the code is well commented.
- B. Tight coupling; changes inside billing can break orders. Use a defined billing interface.
- C. Loose coupling; the modules are too independent.
- D. High cohesion, which is good.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A class named Utilities handles email sending, tax calculation, and PDF generation. What design weakness does it show?

- A. Low cohesion.
- B. Tight coupling.
- C. None; fewer classes are always better.
- D. High cohesion.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A four-person startup with one product and 2,000 users plans to split its app into 25 microservices “because big tech does.” It just closed a seed funding round. What is the best advice?

- A. Start with a well-modularized monolith and split out services only when team or scaling needs demand it.
- B. Build all 25 microservices; they are always better.
- C. Go serverless to avoid architecture decisions.
- D. Use the seed money to hire an operations team for the microservices.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

When do microservices most clearly beat a monolith?

- A.** When components need the lowest possible latency between them.
- B.** When a small team builds a simple app.
- C.** Always, for any system.
- D.** When many teams need to deploy and scale parts of the system independently.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In an app with presentation, business, and data-access layers, a developer puts SQL queries directly into a web page's button-click handler to save time before a demo. The demo is for a hospital board. What is the main problem?

- A.** It is fine because the board will only see the demo.
- B.** Layers must never be crossed in any program, even small scripts.
- C.** Data-access logic sits in the presentation layer, tangling UI changes with database changes.
- D.** It violates the Observer pattern.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A weather app must update phone, web, and watch displays whenever a new reading arrives, and new display types will be added later. The team of seven writes in Java. Which pattern fits?

- A.** Singleton.
- B.** Observer.
- C.** Adapter.
- D.** Factory.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A system wraps an old payment library so its interface matches the one the new app expects. Which pattern is this?

- A.** Observer.
- B.** Adapter.
- C.** Decorator.
- D.** Strategy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A retail site on one large server hits 100% CPU on its biggest sale day. User sessions are stored in that server's memory. The server is three years old. To scale horizontally behind a load balancer, what must change first?

- A. Buy a bigger server.
- B. Add servers; nothing else needs to change.
- C. Move session state to a shared store so any server can handle any request.
- D. Replace the three-year-old server.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Every request passes through a web tier (99.9% available), an app tier (99.9%), and a database (99.5%); all three are required. The database vendor is a market leader. What is the end-to-end availability?

- A. About 99.3%
- B. 99.9%
- C. 99.5%
- D. About 99.77%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Two redundant load balancers are each 99% available, and the system works if either one is up. What is the combined availability?

- A. 99%
- B. 100%
- C. 98.01%
- D. 99.99%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An architect designs a patient portal with a strong perimeter firewall; inside it, all services trust each other and share one administrator database account. The hospital was recently voted a top employer. What is the main design flaw?

- A.** The firewall needs more rules.
- B.** Staff morale is the main risk.
- C.** No defense in depth or least privilege; one compromised service could reach all data. Give each service limited credentials and authenticate internal calls.
- D.** A strong firewall alone is enough.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

When an order is placed, the checkout service calls the email, inventory, and analytics services one after another and waits for each. When analytics is slow, checkout times out. The company processes 10,000 orders a day. What is the best change?

- A.** Cap orders below 10,000 a day.
- B.** Merge all four services into one.
- C.** Raise the checkout timeout.
- D.** Publish an OrderPlaced event to a message queue and let email and analytics consume it asynchronously.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

An order module queries the billing module's internal database tables directly and reads 14 of its private fields. ~~The code is eight years old and well commented.~~ What is the main design problem?

Target word: name the flaw **Decides:** One module depends on another's internals.

A	Camouflage	Comments do not reduce coupling.
B	Correct	Depending on private internals couples the modules.
C	Inverted operation	The opposite of what is happening.
D	Confusion set	Cohesion is about focus within a module, not dependencies between them.

2. Answer A Concept 1 · transfer

A class named Utilities handles email sending, tax calculation, and PDF generation. What design weakness does it show?

Target word: name the weakness — reversed **Decides:** Unrelated jobs in one class.

A	Correct	The class has no single focused purpose.
B	Adjacent category	Coupling is about dependencies between modules.
C	Absolute language	Fewer classes can mean tangled responsibilities.
D	Inverted operation	High cohesion means one focused purpose.

3. Answer A Concept 2

A four-person startup with one product and 2,000 users plans to split its app into 25 microservices “because big tech does.” ~~It just closed a seed funding round.~~ What is the best advice?

Target word: architecture choice **Decides:** Small team, simple product.

A	Correct	Microservice overhead outweighs benefits at this size.
B	Absolute language	They add operational cost that a small team must carry.
C	Adjacent category	Still requires architecture decisions.
D	Camouflage	Funding does not make the design fit.

4. Answer D Concept 2 · transfer

When do microservices most clearly beat a monolith?

Target word: fit — reversed **Decides:** Independent teams and scaling.

A	Adjacent category	Network calls add latency compared with in-process calls.
B	Inverted operation	A monolith is simpler there.
C	Absolute language	They carry real costs.
D	Correct	That is the problem microservices solve.

5. Answer C Concept 3

In an app with presentation, business, and data-access layers, a developer puts SQL queries directly into a web page's button-click handler to save time before a demo. ~~The demo is for a hospital board.~~ What is the main problem?

Target word: layer violation **Decides:** Data access in the presentation layer.

A	Camouflage	The audience does not change the design flaw.
B	Absolute language	The issue is this app's chosen structure, not a universal ban.
C	Correct	Skipping layers breaks separation of concerns.
D	Confusion set	Observer is unrelated to layering.

6. Answer B Concept 4

A weather app must update phone, web, and watch displays whenever a new reading arrives, and new display types will be added later. ~~The team of seven writes in Java.~~ Which pattern fits?

Target word: pattern choice **Decides:** One source notifies many changing subscribers.

A	Confusion set	Ensures one instance; does not notify.
B	Correct	Subscribers register and are notified of changes.
C	Confusion set	Converts interfaces.
D	Adjacent category	Creates display objects but does not push updates.

7. Answer B Concept 4 · transfer

A system wraps an old payment library so its interface matches the one the new app expects. Which pattern is this?

Target word: pattern — reversed from description **Decides:** Interface conversion.

A	Confusion set	Notifies subscribers.
B	Correct	Converts one interface into another.
C	Adjacent category	Adds behavior while keeping the same interface.
D	Inverted operation	Swaps interchangeable algorithms behind one interface rather than converting a mismatched one.

8. Answer C Concept 5

A retail site on one large server hits 100% CPU on its biggest sale day. User sessions are stored in that server's memory. ~~The server is three years old.~~ To scale horizontally behind a load balancer, what must change first?

Target word: prerequisite for horizontal scaling **Decides:** Sessions tied to one server's memory.

A	Adjacent category	Vertical scaling, not horizontal.
B	Dropped element	Users lose sessions when routed to another server.
C	Correct	Stateless app servers can be added freely.
D	Camouflage	Age does not decide the scaling blocker.

9. Answer A Concept 6

Every request passes through a web tier (99.9% available), an app tier (99.9%), and a database (99.5%); all three are required. ~~The database vendor is a market leader.~~ What is the end-to-end availability?

Target word: series availability **Decides:** Multiply the three.

A	Correct	$0.999 \times 0.999 \times 0.995 \approx 0.9930$.
B	Wrong stem number	One tier's figure.
C	Dropped element	Uses only the weakest link.
D	Adjacent category	Averages the three.

10. Answer D Concept 6 · transfer

Two redundant load balancers are each 99% available, and the system works if either one is up. What is the combined availability?

Target word: parallel availability **Decides:** $1 - (0.01 \times 0.01)$.

A	Dropped element	Ignores the redundancy.
B	Absolute language	Both can still fail at once.
C	Inverted operation	Multiplies as if in series.
D	Correct	Both fail together only 0.01% of the time.

11. Answer C Concept 7

An architect designs a patient portal with a strong perimeter firewall; inside it, all services trust each other and share one administrator database account. ~~The hospital was recently voted a top employer.~~ What is the main design flaw?

Target word: design-level security flaw **Decides:** One shared admin account and implicit internal trust.

A	Adjacent category	Strengthens the perimeter but leaves the flat trust inside.
B	Camouflage	The award does not decide the architecture flaw.
C	Correct	Layered controls limit how far a single failure spreads.
D	Absolute language	A single barrier is a single point of failure.

12. Answer D Concept 8

When an order is placed, the checkout service calls the email, inventory, and analytics services one after another and waits for each. When analytics is slow, checkout times out. ~~The company processes 10,000 orders a day.~~ What is the best change?

Target word: decouple a slow dependency **Decides:** Checkout waits on non-critical services.

A	Camouflage	Volume is not the design flaw.
B	Inverted operation	Increases coupling.
C	Dropped element	Customers still wait; the coupling remains.
D	Correct	Checkout no longer waits on slow consumers.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Coupling and cohesion	Aim for low coupling (modules depend on each other only through clear interfaces) and high cohesion (each module has one focused purpose).	Mixing up the two, or calling reaching into another module's internals "integration."	A module reading another's private data, or a grab-bag "Utilities" class.
2	Monolith vs. microservices	Microservices pay off when many teams must deploy and scale parts independently; they add network and operations overhead.	Adopting microservices by default for small teams and simple products.	Team size, product scope, and "because big companies do."
3	Layered architecture	Presentation calls business logic, which calls data access; skipping layers entangles concerns.	Putting data queries in UI code to save time.	SQL in a button handler.
4	Design patterns	Observer notifies many subscribers of changes; Adapter converts one interface to another; Strategy swaps algorithms; Factory creates objects; Singleton ensures one instance.	Confusing patterns with similar-sounding purposes.	"Notify all displays" or "wrap the old interface."
5	Horizontal scaling needs stateless services	To add servers behind a load balancer, move session state out of each server into a shared store.	Adding servers while sessions live in local memory, or only buying a bigger box.	Sessions stored in server memory and a traffic spike.
6	Availability math	Components in series: multiply availabilities. Redundant components in parallel: 1 – product of their unavailabilities.	Averaging, or treating the weakest link as the total.	Several percentages and "all required" vs. "either works."
7	Secure by design	Use defense in depth and least privilege: segment, authenticate internal calls, and give each service its own limited credentials.	Trusting everything inside a strong perimeter.	One shared admin account and "inside the firewall is trusted."
8	Asynchronous messaging	Publishing events to a queue decouples producers from slow consumers and improves resilience.	Raising timeouts or merging services.	One slow downstream call making the main request time out.

Course scope (public description): This course covers topics in designing, analyzing, and managing large-scale software systems. Students will learn various architecture types, how to select and implement appropriate design patterns, and how to build well-structured, reliable, and secure software systems.